Concyclic Euclidean Ramsey sets with three-dimensional moduli

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October 23, 2026

Abstract

Let A be a finite set of at least seven points on a circle, and let M be the field generated over $\mathbb{Q}$ by its four-point cross ratios. When $\operatorname{trdeg}_{\mathbb{Q}} M = 3$, the set is Euclidean Ramsey precisely when it is a union of two groups, each containing at least three points and having only algebraic internal cross ratios. For seven points this is equivalent to the existence of an algebraic four-point cross ratio. We prove the necessary implication by a second-order tensor obstruction and a classification of simultaneous totally isotropic three-dimensional subspaces. The sufficient implication is an application of OpenAI's Theorem 1.1 and is also covered by Strinz's two-group theorem. The criterion decides the three-parameter example left open in Appendix E of Pálvölgyi's paper: it is non-Ramsey. This negative conclusion, and the necessary implication above, use neither direction of OpenAI's classification; the needed tensor necessity is proved here. An explicit elliptic family, whose Ramsey conclusion uses the sufficient direction of that classification, has one-dimensional moduli and no algebraic four-point cross ratio. It illustrates the role of the dimension hypothesis; other examples of this phenomenon are already public.

1 Statements and background

A finite nonempty set A in a Euclidean space is *Ramsey* if, for every integer $r \geq 2$, there exists $N \geq 1$ such that every colouring $\mathbb{R}^N \to \{1, \dots, r\}$ contains a monochromatic congruent copy of A. The scale is fixed, and the colouring has no regularity assumption. Ramsey-ness depends only on the distance table, is preserved by nonzero similarities, and passes to subsets. The assertions concerning circle sets below apply to circles of positive radius in any Euclidean space: we first choose an isometric planar representative and then use a similarity to the unit circle.

Use homogeneous parameters $z = [s : t] \in \mathbb{P}^1(\mathbb{R})$ for the unit circle:

$$a([s : t]) = \left(\frac{s^2 - t^2}{s^2 + t^2}, \frac{2st}{s^2 + t^2} \right). \tag{1.1}$$

For nonzero representatives put $[z_i, z_j] = s_i t_j - t_i s_j$. For four distinct indices define

$$\operatorname{cr}(i, j; k, l) = \frac{[z_i, z_k][z_j, z_l]}{[z_i, z_l][z_j, z_k]}. \tag{1.2}$$

All denominators are nonzero. This quantity is unchanged by rescaling representatives or applying a common element of $\mathrm{PGL}_2(\mathbb{R})$. Changing the projective parametrization of the circle therefore does not change the following field.

Definition 1.1. The *projective moduli field* of a labelled circle set A is

$$M(A) = \mathbb{Q}\big(\mathrm{cr}(i,j;k,l) : i,j,k,l \text{ distinct}\big).$$

A group of at least three of its points is *projectively algebraic* if all its internal four-point cross ratios are algebraic over $\mathbb{Q}$. For a group of three points this condition is vacuous.

Equivalently, a projectively algebraic group is a projective image of a finite subset of $\mathbb{P}^1(\overline{\mathbb{Q}} \cap \mathbb{R})$. Indeed, sending three of its points to $0, 1, \infty$ makes every remaining parameter a cross ratio or a fractional linear transform of one. The same observation shows that $M(A)$ is generated by the remaining $n-3$ parameters after such a normalization. In particular, the moduli degree refers to the normalized parameters, rather than the absolute coordinate field.

Theorem A. *Let A consist of $n \geq 7$ distinct points on a circle and suppose that $\mathrm{trdeg}_{\mathbb{Q}}\, M(A) = 3$. Then the following are equivalent:*

1. *A is Ramsey;*
2. *its indices have a partition $I \sqcup J$, with $|I|, |J| \geq 3$, such that each group is projectively algebraic.*

The implication (2)⇒(1) uses the sufficient direction of OpenAI's Theorem 1.1 [2]; it is also a circle case of Strinz's two-group theorem [6, Theorem C2]. The implication (1)⇒(2) is proved here without either direction of OpenAI's theorem.

Corollary B. *For seven distinct circle points with $\mathrm{trdeg}_{\mathbb{Q}}\, M(A) = 3$, the set is Ramsey if and only if some four-point cross ratio is algebraic over $\mathbb{Q}$. The sufficient implication uses OpenAI's Theorem 1.1, sufficient direction; the necessary implication uses neither direction.*

In a partition of seven indices into groups of size at least three the sizes are 3 and 4. For a four-point group all orderings of its cross ratio are fractional linear transforms of one of them. Thus the equivalence in Corollary B follows immediately from Theorem A. Its sufficient implication holds without a moduli-degree hypothesis, as already follows from Strinz's results.

Corollary C. *Let x, y, z be algebraically independent real numbers over $\mathbb{Q}$. The seven circle points with finite parameters*

$$-2, \quad 0, \quad 2, \quad x, \quad y, \quad -4-x-2y, \quad z \tag{1.3}$$

are non-Ramsey. This conclusion uses neither direction of OpenAI's Theorem 1.1.

This is the family discussed after Theorem A.8 in Appendix E of Pálvölgyi [4]. His parameters are $s_1 = x$, $s_2 = y$, $s_3 = -4 - s_1 - 2s_2$, and $s_4 = z$. That discussion leaves the Ramsey status open and imposes a neighbourhood condition for the failure of its derivation-certificate method. The corollary above has no neighbourhood condition. Section 6 gives a short cross-ratio proof and an independent finite second-order certificate.

Theorem D. *For each transcendental real $X > \sqrt[3]{2}$ and either choice $Y = \pm\sqrt{X^3 - 2}$, the recurrence in (7.1) produces seven distinct circle parameters $X_0, \ldots, X_6$. Their circle set is Ramsey, its moduli field has transcendence degree one, and all its four-point cross ratios are transcendental. The Ramsey assertion uses the sufficient direction of OpenAI's Theorem 1.1.*

This example shows that the degree-three hypothesis in the necessary implication of Corollary B cannot simply be omitted. The existence of Ramsey seven-point sets with no algebraic cross ratio is already exhibited by Strinz [6, Section 7, "Non-two-group Ramsey sets"], using different and simpler rational families, including

$$\{0, \tfrac{1}{2}, \tfrac{1}{3}, t, t+1, 2t, 1/t\} \qquad (t \text{ real and transcendental}).$$

That positive conclusion also uses the sufficient direction of OpenAI's Theorem 1.1. The elliptic example is included as a concrete boundary construction.

Public results used for context

OpenAI's *A classification of finite Euclidean Ramsey configurations*, dated September 23, 2026, states a necessary and sufficient tensor condition in Theorem 1.1 [2]. We use the release at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`, result 172. The corresponding Lean development includes `fieldCriterion_ramsey` and `classification` in `OAI/Combinatorics/EuclideanRamsey/Main.lean` at that commit [3]. Its source supplies both directions; no Lean kernel replay is part of the present note.

Jones's October 7, 2026 note gives an explicit tensor certificate for any six concyclic points [1, Section 2]. Composed with the sufficient direction of OpenAI's Theorem 1.1, this proves that every circle set with at most six points is Ramsey. Strinz's October 15, 2026 release states this conclusion as Theorem B(a) and supplies a Lean development composing the six-point construction with the sufficient direction of OpenAI's Theorem 1.1 [6]. Thus the six-point result is background here.

Pálvölgyi's September 20, 2026 preprint states in Theorem A.3 that, for any fixed distinct real t_1, t_2, t_3, if $t_4, \dots, t_7$ are algebraically independent over $\mathbb{Q}(t_1, t_2, t_3)$, the seven circle points are non-Ramsey [4]. That negative result has a derivation and colouring proof and uses neither direction of OpenAI's Theorem 1.1. Its applicability is stronger than independence of all seven absolute parameters, but does not cover (1.3).

Strinz's Möbius-invariance theorem, released October 13, 2026, states that a finite spherical set is Ramsey if and only if its image under a Möbius transformation of its containing sphere is Ramsey [5, Theorem C]. Its published argument uses both directions of OpenAI's Theorem 1.1. The October 15 spherical note gives two further sufficient conditions: two Möbius images of finite algebraic subsets of a sphere have Ramsey union (Theorem C2), and an arbitrary finite algebraic circle set together with at most three arbitrary circle points is Ramsey (Theorem B(b)) [6]. Both positive conclusions use the sufficient direction of OpenAI's Theorem 1.1. Their circle case explains the sufficient halves of Theorem A and Corollary B. The explicit certificate is recalled in Section 5.2.

For comparison, on S^{d-1} with $d \geq 3$, algebraically independent stereographic parameters give the generic threshold $\binom{d+2}{2}$: fewer points are Ramsey and that many or more are non-Ramsey. Strinz's ledger SR-C3 and SR-C29 records the positive half as a consequence of OpenAI's Proposition 7.3, using the sufficient direction of its Theorem 1.1; the negative boundary is Pálvölgyi's Theorem A.12 and uses neither direction [6, 4]. On the circle, the relevant generic negative size is seven, and the classification below concerns the degree-three locus within its moduli.

The proofs occupy Sections 2–5. Section 6 treats Corollary C, and Section 7 proves Theorem D. The ancillary files record exact finite checks; the structural necessity argument itself is given over the full moduli field.

2 The tensor criterion and its necessary direction

For a field $F \subset \mathbb{R}$ write

$$\mathcal{B}_F = F \otimes_{\mathbb{Q}} F, \qquad L(x) = x \otimes 1, \qquad R(x) = 1 \otimes x, \qquad m(x \otimes y) = xy.$$

Every nonzero pure factor $L(x)$ or $R(x)$ is a unit. The ring $\mathcal{B}_F$ need not be a domain. If $a_1, \dots, a_n \in F^d$, put $p_i = (1, a_i)^T$. A *tensor certificate* is a matrix $P \in \mathrm{Mat}_{d+1}(\mathcal{B}_F)$ with

$$L(p_i)^T P R(p_i) = 0 \quad (1 \leq i \leq n), \qquad m(P_{\alpha\beta}) = \delta_{\alpha\beta} \quad (1 \leq \alpha, \beta \leq d). \tag{2.1}$$

The constant row and column have no prescribed multiplied values. OpenAI's Theorem 1.1 says that a distinct configuration of at least two points spanning $\mathbb{R}^d$ is Ramsey if and only

if (2.1) holds in its coordinate field [2]. The sufficient direction is the input for all positive conclusions of this note. To specify the dependence of the negative conclusions, we give the elementary necessity argument separately; it is the tensor separation and colouring mechanism of [2, Section 2], with the required descent made explicit.

Lemma 2.1 (Descent)**.** *Let $F \subset E \subset \mathbb{R}$, and suppose the coordinates lie in F. A tensor certificate over $E \otimes_{\mathbb{Q}} E$ descends to one over $F \otimes_{\mathbb{Q}} F$. The same assertion holds if the whole multiplied matrix is prescribed to be a fixed matrix over F.*

Proof. Choose an F-basis (e_λ) of E. There is a free-module decomposition

$$E \otimes_{\mathbb{Q}} E = \bigoplus_{\lambda,\mu} (F \otimes_{\mathbb{Q}} F)(e_\lambda \otimes e_\mu).$$

Expand the given matrix as a finite sum $P_E = \sum (e_\lambda \otimes e_\mu) P_{\lambda\mu}$. Since its evaluation coefficients lie in $F \otimes_{\mathbb{Q}} F$, each $P_{\lambda\mu}$ has zero evaluations. If $H_{\lambda\mu} = m(P_{\lambda\mu})$, the required entries of the target matrix H satisfy

$$H = \sum_{\lambda,\mu} e_\lambda e_\mu H_{\lambda\mu},$$

where only the prescribed entries are included if appropriate. This is a finite linear system over F, soluble over E. Row reduction over F shows that it has a solution $c_{\lambda\mu} \in F$. Then $\sum L(c_{\lambda\mu}) P_{\lambda\mu}$ is the required certificate. □

Proposition 2.2 (Independent necessity)**.** *If a distinct, affinely spanning configuration $a_1, \dots, a_n \in \mathbb{R}^d$ has no tensor certificate in its coordinate field, it is non-Ramsey. In every ambient dimension there is an avoiding colouring with at most $(2n+1)^n$ colours. This statement uses neither direction of OpenAI's Theorem 1.1.*

Proof. Descent permits us to work with $\mathbb{R} \otimes_{\mathbb{Q}} \mathbb{R}$. Regard $W_* = \mathbb{R} \times \mathbb{R}^d$ as a $\mathbb{Q}$-vector space, and write $e_i(v, u) = v + a_i \cdot u$. On the subspace $T(W_*)$ of symmetric tensors in $W_* \otimes_{\mathbb{Q}} W_*$ define

$$\Psi(T) = \big((e_i \otimes e_i)(T)\big)_{i=1}^n \oplus G(T), \qquad G\big((v,u) \otimes (v',u')\big) = u(u')^T.$$

Here $G(T)$ is a real symmetric matrix. Distributing the finite coordinate sums identifies these tensors with matrices over $\mathbb{R} \otimes_{\mathbb{Q}} \mathbb{R}$. Under this identification, the first components are the tensor evaluations and the last component is the multiplied spatial block. A certificate can always be made symmetric for the combined transpose and tensor flip by averaging. Hence $y_* = (0, \dots, 0, I_d)$ is outside the image of Ψ.

Separation in the quotient $\mathbb{Q}$-vector space gives a $\mathbb{Q}$-linear functional vanishing on $\operatorname{im}\Psi$ and taking y_* to one: choose a basis containing the nonzero class of y_*. Write its components as $\ell_i : \mathbb{R} \otimes_{\mathbb{Q}} \mathbb{R} \to \mathbb{R}$ and $\Lambda : \operatorname{Sym}_d(\mathbb{R}) \to \mathbb{R}$, and put $h_i(x) = \ell_i(x \otimes x)$. On a pure square this gives

$$\sum_i h_i(v + a_i \cdot u) + \Lambda(uu^T) = 0, \qquad \Lambda(I_d) = 1. \tag{2.2}$$

Taking $u = 0$ shows $\sum_i h_i(x) = 0$ for all real x.

In $\mathbb{R}^N$ define $H_i(z) = \sum_{j=1}^N h_i(z_j)$. Every congruent copy has the form $b_i = t + Aa_i$ with $A^T A = I_d$: the distance identities identify the Gram matrices of the spanning difference vectors, giving a linear isometric embedding and then a translation. If $N < d$ there is no such copy. Apply (2.2) to each coordinate row of A and add. Additivity of Λ gives

$$\sum_i H_i(z) = 0 \quad (z \in \mathbb{R}^N), \qquad \sum_i H_i(b_i) = -\Lambda(A^T A) = -1.$$

No real scalar is pulled through the merely $\mathbb{Q}$-linear functional.

Partition the residues modulo 2 of each $H_i(z)$ into $2n+1$ equal half-open intervals of $[0,2)$. Their n labels colour z. In a monochromatic copy, $H_i(b_i) - H_i(b_1) = 2k_i + e_i$ with $k_i \in \mathbb{Z}$ and $|e_i| < 2/(2n+1)$. Summing gives $-1 = 2\sum_i k_i + \sum_i e_i$, where $|\sum_i e_i| < 2n/(2n+1) < 1$. An odd integer cannot have absolute value less than one. This contradiction proves the assertion. □

2.1 Full normalization on the unit circle

Put

$$h(s,t) = (s^2+t^2, s^2-t^2, 2st)^T, \qquad H_0 = \mathrm{diag}(-1,1,1).$$

The components of h form a basis of the binary quadratic forms over any $\mathbb{Q}$-algebra, and

$$h(s,t)^T H_0 h(u,v) = -2(sv-tu)^2. \tag{2.3}$$

Every bihomogeneous polynomial of bidegree $(2,2)$ over $\mathcal{B}_F$ therefore has a unique representation $h(s,t)^T P h(u,v)$. For finite parameters, $h(1,t)^T P h(1,u)$ similarly represents every polynomial of degree at most two in each variable. These are changes of basis in free modules and remain valid when $\mathcal{B}_F$ has zero divisors.

If at least three distinct unit circle points have a certificate, replace it by $(P+P^*)/2$, where $P^*_{\alpha\beta} = \tau(P_{\beta\alpha})$ and $\tau(x \otimes y) = y \otimes x$. Its multiplied matrix is symmetric. Ordinary evaluation of (2.1) gives

$$m(P_{00}) + 1 + 2m(P_{01})a_{i1} + 2m(P_{02})a_{i2} = 0.$$

Three distinct circle points are not collinear. This affine function must vanish identically, whence

$$m(P) = H_0. \tag{2.4}$$

Thus prescribing the full matrix here loses no possible certificate.

2.2 Projective transport without a Ramsey input

For $A \in \mathrm{GL}_2(\mathbb{R})$, let C_A be its action on binary quadratic forms, so $h(Az) = C_A h(z)$. Equation (2.3) and $[Az, Aw] = \det(A)[z,w]$ give

$$C_A^T H_0 C_A = \kappa H_0, \qquad \kappa = (\det A)^2.$$

In a common real field containing the coordinates and the entries of A, a certificate with (2.4) transports to

$$P' = L(\kappa) L(C_A^{-T}) P R(C_A^{-1}). \tag{2.5}$$

Indeed the new augmented circle vectors are $C_A p_i$ divided by their constant coordinates. These coordinates equal $\|Az_i\|^2/\|z_i\|^2 > 0$ and are units in the pure factors. The evaluations therefore vanish and $m(P') = H_0$. Lemma 2.1 descends P' to the image coordinate field. The inverse transformation gives the converse. Thus the existence of a tensor certificate for a circle set of at least three points is projectively invariant by algebra alone. When this fact is used to prove a positive Ramsey conclusion, the sufficient direction of OpenAI's Theorem 1.1 is still required. Its use in a negative proof requires only Proposition 2.2.

3 A second-order condition over the moduli field

Normalize three distinct parameters to $0, 1, \infty$. Choose a rational α different from every finite normalized parameter, including 0 and 1, and apply $r \mapsto 1/(r-\alpha)$. Every parameter is now finite and the first three are distinct rational numbers. This operation and its inverse are rational over $\mathbb{Q}$, so the field of all finite parameters remains the moduli field $F = M(A)$. The unit circle coordinates have this same field, since

$$a(t) = \left(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2}\right), \qquad t = \frac{a_2}{1+a_1}.$$

The denominator in the inverse is $2/(1+t^2) > 0$. Section 2.2 justifies using this normalization for certificate existence without a Ramsey-invariance input.

Write the distinct finite parameters as $t_1, \dots, t_n$ and let $k = \operatorname{trdeg}_{\mathbb{Q}} F$. Choose a transcendence basis $u_1, \dots, u_k$. The field F is finite separable over $\mathbb{Q}(u_1, \dots, u_k)$. Each partial derivation extends uniquely to F: for an algebraic element β with minimal polynomial f, differentiating $f(\beta) = 0$ prescribes

$$D(\beta) = -\frac{(Df)(\beta)}{f'(\beta)}.$$

Here Df differentiates the coefficients. Separability makes the denominator nonzero, and the prescription preserves the defining ideal of the simple extension. Applying this to finite simple extensions proves existence and uniqueness. It follows that $\operatorname{Der}_{\mathbb{Q}}(F, F)$ has basis $D_1, \dots, D_k$, where $D_j(u_l) = \delta_{jl}$. Its common constants are precisely the elements of F algebraic over $\mathbb{Q}$: a transcendental element can itself be included in a transcendence basis and assigned derivative one.

Let

$$V = \big\{(D(t_i))_{i=1}^n : D \in \operatorname{Der}_{\mathbb{Q}}(F, F)\big\}, \qquad W = \big\{(q(t_i))_{i=1}^n : \deg q \le 2\big\}. \tag{3.1}$$

Since the t_i generate F, the velocity map is injective, and $\dim_F V = k$. Its first three entries vanish. A quadratic vanishing at those three distinct parameters is zero, so

$$V \cap W = 0, \qquad \dim W = 3. \tag{3.2}$$

Set

$$H(T) = \prod_{i=1}^{n} (T - t_i), \qquad \gamma_i = H'(t_i)^{-1}. \tag{3.3}$$

Lagrange interpolation, followed by comparison of the coefficient of T^{n-1}, gives

$$\sum_i \gamma_i t_i^r = 0 \quad (0 \le r \le n-2), \qquad \sum_i \gamma_i t_i^{n-1} = 1. \tag{3.4}$$

Proposition 3.1 (The graph condition)**.** *Suppose $n \ge 7$ and the circle set has a tensor certificate. There is an F-linear map $Q : V \to W$ such that*

$$B_l(v, Qw) + B_l(w, Qv) = 4B_l(v, w) \quad (v, w \in V,\ 0 \le l \le n-6), \tag{3.5}$$

where $B_l(a, b) = \sum_i \gamma_i t_i^l a_i b_i$. Consequently

$$U = \{v - \tfrac{1}{4} Qv : v \in V\} \tag{3.6}$$

has dimension k, is totally isotropic for every B_l, and satisfies $V \subset U + W$.

Proof. Normalize the certificate by (2.4). In the truncated ring

$$\mathcal{R} = F[\delta_1, \dots, \delta_k]/(\delta_1, \dots, \delta_k)^3$$

put $D_\delta = \sum_j \delta_j D_j$, acting trivially on the δ_j. The map

$$\varphi(x) = x + D_\delta x + \tfrac{1}{2} D_\delta^2 x$$

is a $\mathbb{Q}$-algebra homomorphism. This follows by applying Leibniz twice and discarding terms of degree at least three. In particular it respects inverses, since a nonzero constant term is a unit. Commutation of the individual derivations is not needed for this calculation. Map the left factor of $\mathcal{B}_F$ identically to F and the right factor by φ.

Using $h(1,t)$, the image matrix represents a polynomial

$$\mathcal{G}(t,u) = -2(t-u)^2 + \mathcal{G}_1(t,u) + \mathcal{G}_2(t,u),$$

of degree at most two in each variable, with $\mathcal{G}_j$ homogeneous of degree j in δ. The first-order node equations give $\mathcal{G}_1(t_i,t_i) = 0$. Each coefficient on the diagonal is a polynomial of degree at most four with at least seven roots; hence $\mathcal{G}_1(t,t) = 0$. Polynomial division gives

$$\mathcal{G}_1(t,u) = (u-t)N(t,u), \qquad N(t,t) = \sum_j a_j(t)\delta_j, \quad \deg a_j \le 2.$$

The quotient N has degree at most one in each variable: multiplication by $u-t$ raises the degree in either variable by one. Put $q(t) = \mathcal{G}_2(t,t)$, a polynomial in t of degree at most four. With $v_i = \sum_j D_j(t_i)\delta_j$, the second-order node equations are

$$q(t_i) = 2v_i^2 - v_i \sum_j a_j(t_i)\delta_j. \tag{3.7}$$

The quadratic displacement in $\varphi(t_i)$ contributes no term of order two: the leading kernel has a double zero on the diagonal, and $\mathcal{G}_1$ already has order one in δ.

For a velocity $v = \sum_j c_j(D_j(t_i))_i$, define $Qv = (\sum_j c_j a_j(t_i))_i$. This is well-defined because the velocity columns are independent, and it takes values in W. Multiplying (3.7) by $\gamma_i t_i^l$ and summing, (3.4) gives $B_l(v, Qv) = 2B_l(v,v)$ for $0 \le l \le n-6$. Polarization proves (3.5). Also (3.4) makes W totally isotropic for every such B_l, because $l+4 \le n-2$. Expanding $B_l(v - Qv/4, w - Qw/4)$ now gives zero. The graph map is injective by (3.2), so $\dim U = k$; its definition gives $V \subset U + W$. □

There is a useful equivalent form of this condition. The $(n-5) \times n$ matrix $(\gamma_i t_i^l)_{0 \le l \le n-6}$ has rank $n-5$ by the Vandermonde determinant. Its kernel is exactly the five-dimensional space of values of polynomials of degree at most four, by (3.4). Thus

$$u, w \in U \quad \Longrightarrow \quad (u_i w_i)_{i=1}^n \text{ is the value vector of a polynomial of degree } \le 4. \tag{3.8}$$

This formulation will allow restriction to eight nodes.

4 Common isotropic three-dimensional subspaces

The linear algebra in this section is over an arbitrary field of characteristic zero. Nodes are distinct, γ_i is defined by (3.3), and W is the quadratic evaluation space. A subspace is *totally isotropic* for a symmetric bilinear form if that form vanishes on every pair of its vectors.

Lemma 4.1 (Seven nodes)**.** *For seven nodes, every three-dimensional subspace totally isotropic for B_0 and B_1 is*

$$U = EW, \qquad E = \operatorname{diag}(\varepsilon_1, \dots, \varepsilon_7), \quad \varepsilon_i \in \{1, -1\}. \tag{4.1}$$

There are exactly 2^6 such spaces.

Proof. Let $T = \operatorname{diag}(t_i)$. Since $B_1(a, b) = B_0(Ta, b)$, we have $TU \subset U^{\perp_{B_0}}$. The latter has dimension four and contains U. Thus $K = \{u \in U : Tu \in U\}$ has dimension at least two. The linear map $K \to U/K$, $u \mapsto Tu \bmod K$, has a nonzero kernel. Choose $u \neq 0$ in it; then $u, Tu, T^2u \in U$.

These vectors are independent. Otherwise $f(T)u = 0$ for some nonzero polynomial of degree at most two, so u is supported on at most two nodes. But $B_0(u, u) = B_1(u, u) = 0$ is an invertible Vandermonde system on those nodes for the quantities $\gamma_i u_i^2$. It would force $u = 0$.

The three vectors therefore form a basis of U. Applying both forms to their pairs yields

$$\sum_i \gamma_i t_i^r u_i^2 = 0 \qquad (0 \le r \le 5).$$

The 6×7 Vandermonde matrix has a one-dimensional kernel, spanned by $(\gamma_i)_i$ by (3.4). Hence $u_i^2 = c$ for every i. Since $u \neq 0$, $c \neq 0$. Dividing by u_1 gives $u_i/u_1 = \varepsilon_i \in \{1, -1\}$ and proves (4.1). Conversely every such space is isotropic by (3.4).

For the count, suppose a relative sign matrix preserves W. Its sign vector is the evaluation of a polynomial q of degree at most two, and its product with $(t_i^2)_i$ is the evaluation of another such polynomial r. The polynomial $T^2 q(T) - r(T)$ has degree at most four and seven roots, so is zero. Thus q is constant, and all relative signs coincide. Only a common sign is redundant, giving 2^6 spaces. □

Lemma 4.2 (Eight nodes)**.** *For eight nodes, a three-dimensional subspace totally isotropic for B_0, B_1, B_2 is EW for a sign matrix E.*

Proof. The map T is injective on U. Indeed a vector in its kernel would be supported at the unique possible zero node, and a nonzero such vector has $B_0(u, u) \neq 0$. The subspace $U + TU$ is totally isotropic for B_0: the U, U pair uses B_0, the mixed pair uses B_1, and the TU, TU pair uses B_2. Nondegeneracy of B_0 in dimension eight gives $\dim(U + TU) \le 4$. Since $\dim TU = 3$, the space $K = \{u \in U : Tu \in U\}$ again has dimension at least two. The preceding kernel argument supplies $u, Tu, T^2u \in U$; their independence follows as in Lemma 4.1. The three forms on these vectors give

$$\sum_i \gamma_i t_i^r u_i^2 = 0 \qquad (0 \le r \le 6).$$

The 7×8 Vandermonde kernel is spanned by $(\gamma_i)_i$, so all u_i^2 are equal and nonzero. Normalization gives $U = EW$. The sign-uniqueness argument in Lemma 4.1 also applies to eight nodes; there are 2^7 such spaces. □

Proposition 4.3 (Restriction and gluing)**.** *Let $n \ge 7$. Suppose $U \subset F^n$ has dimension three and satisfies the product property* (3.8)*. Then $U = EW$ for a sign matrix E.*

Proof. For $n = 7$, the product property implies isotropy for B_0, B_1 by interpolation, so use Lemma 4.1. Suppose $n \ge 8$. Choose eight coordinate positions such that restriction of U has rank three: choose three independent coordinate rows of a basis matrix and add five positions. The restricted product vectors are quartic evaluations. Interpolation on these eight nodes

makes the restricted space isotropic for its own B_0, B_1, B_2. Lemma 4.2 makes it a signed quadratic evaluation space. In particular, any three of its coordinate rows are independent.

For any additional node, combine it with any seven of the chosen eight. The new restriction has rank three and is again a signed quadratic space. On the common seven nodes the two spaces agree: both are the image of U there, and restriction has rank three. The uniqueness of signs in Lemma 4.1 makes their signs agree up to one common reversal. After this reversal, a fixed vector of U is represented on both sets by quadratic polynomials agreeing at seven nodes; they are the same polynomial. Thus the new sign extends the original signs consistently for every vector in U. Repeating for each remaining node proves $U = EW$ globally. □

Lemma 4.4 (The two sign groups)**.** *Suppose $n \geq 7$, $\dim V = 3$, $V \cap W = 0$, and $V \subset W + EW$. Then each sign occurs at least three times. Moreover, restriction of every vector of V to either sign group is the evaluation of a quadratic polynomial on that group.*

Proof. Let m be the size of the smaller sign group I and let J be the other group. If $m \leq 2$, then $|J| \geq 5$. For a vector in $W \cap EW$, write its values as $r(t_i)$ and $\varepsilon_i q(t_i)$ and reverse all signs so that $\varepsilon_i = 1$ on J. On J the quadratics r, q agree, hence are identical. On I the polynomial q must vanish. Therefore $\dim(W \cap EW) = 3 - m$ and

$$\dim((W + EW)/W) = m < 3.$$

This contradicts the injective image of V in the quotient. Both groups consequently have size at least three. Finally, an element of $W + EW$ has the form $(p(t_i) + \varepsilon_i q(t_i))_i$. Its restrictions are the quadratics $p + q$ and $p - q$, respectively. □

All these classifications are field arguments, rather than enumerations of rational examples. As arithmetic checks, `structure.py` verifies the 64 seven-node sign spaces and their projection dimensions, and `eight_factory.py` checks the 128 eight-node sign spaces and $\dim(U+TU) = 4$ on distinct rational nodes. These checks supplement the proofs above.

5 Proof of the classification

5.1 Necessity

Suppose the circle set in Theorem A is Ramsey. Proposition 2.2 gives a tensor certificate. Transport and descent give a certificate for the finite rational-frame normalization of Section 3. In its moduli field, $\dim V = 3$ and $V \cap W = 0$. Proposition 3.1 provides a three-dimensional space U satisfying the product property, with $V \subset U + W$. Proposition 4.3 gives $U = EW$, and Lemma 4.4 partitions the nodes into two groups of size at least three, on each of which every derivation has quadratic parameter velocity.

For four distinct finite parameters in one group put

$$\rho = \frac{(t_a - t_c)(t_b - t_d)}{(t_a - t_d)(t_b - t_c)}.$$

If $v_i = D(t_i)$, the quotient rule gives

$$\frac{D\rho}{\rho} = \frac{v_a - v_c}{t_a - t_c} + \frac{v_b - v_d}{t_b - t_d} - \frac{v_a - v_d}{t_a - t_d} - \frac{v_b - v_c}{t_b - t_c}. \tag{5.1}$$

For quadratic velocities $v_i = a + bt_i + ct_i^2$, each difference quotient is $b + c(t_i + t_j)$, so the four terms cancel. Thus every $\mathbb{Q}$-derivation of F kills ρ. The constants characterization in Section 3 makes ρ algebraic over $\mathbb{Q}$. All internal cross ratios in both groups are algebraic. This proves the necessary implication using neither direction of OpenAI's Theorem 1.1.

5.2 The known sufficient mechanism

For completeness we record the circle certificate behind the sufficient implication. This is a case of Strinz's two-group theorem [6, Theorem C2]; the conclusion that the certificate implies Ramsey is OpenAI's sufficient direction.

In the unit circle coordinate field F, choose $A_I, A_J \in \mathrm{GL}_2(F)$ taking three points of each group to $0, 1, \infty$. The other normalized parameters are real algebraic elements of F. Choose algebraic homogeneous representatives w_i, so $A_H z_i = \lambda_i w_i$ with $\lambda_i \in F^\times$ for $i \in H$, $H = I, J$.

There is an element $e \in \mathcal{B}_F$ such that

$$m(e) = 1, \qquad e(L(\alpha) - R(\alpha)) = 0 \tag{5.2}$$

for every coordinate α of the finitely many w_i. Here is an explicit construction. If $f_\alpha \in \mathbb{Q}[X]$ is the minimal polynomial, put

$$g_\alpha(X) = \frac{f_\alpha(X)}{X - \alpha} \in F[X], \qquad e_\alpha = L(f'_\alpha(\alpha)^{-1}) g^L_\alpha(R(\alpha)), \qquad e = \prod_\alpha e_\alpha.$$

The superscript L embeds the coefficients in the left factor. The identities $(R(\alpha) - L(\alpha))g^L_\alpha(R(\alpha)) = 0$ and $g_\alpha(\alpha) = f'_\alpha(\alpha) \neq 0$ prove (5.2).

For $z = (s, t)^T$, $w = (u, v)^T$, define the bilinear graph forms

$$K_H(z, w) = \det\big(L(A_H)z, R(A_H)w\big) \qquad (H = I, J).$$

At a node of group H, (5.2) gives $eK_H(L(z_i), R(z_i)) = 0$. Multiplication of coefficients gives $m(K_H) = \det(A_H)(sv - tu)$. Hence

$$G(z, w) = -2L((\det A_I \det A_J)^{-1}) e K_I(z, w) K_J(z, w) \tag{5.3}$$

vanishes at every tensor node, and $m(G) = -2(sv - tu)^2$. The basis identity (2.3) yields a matrix with $m(P) = H_0$. Dividing the node vectors by $s_i^2 + t_i^2 > 0$ gives (2.1). All denominators are nonzero elements of F in pure tensor factors; no domain assumption is used. At least three distinct circle points span the plane, so OpenAI's Theorem 1.1, sufficient direction, applies. This proves Theorem A and hence Corollary B.

In particular, the construction applies to a group of four points with algebraic cross ratio and any remaining triple, without a dimension restriction. More generally it includes the algebraic-plus- three conclusion of Strinz. Both are positive Ramsey consequences of the sufficient direction of OpenAI's Theorem 1.1. The scripts `structure.py` and `eight_factory.py` check the diagonal and graph identities used by these constructions.

6 The example in Pálvölgyi's Appendix E

Proof of Corollary C. Write $w = -4 - x - 2y$. Algebraic independence makes the seven parameters in (1.3) distinct. The three rational anchors $-2, 0, 2$ give a normalization defined over $\mathbb{Q}$ and with rational inverse. Therefore the moduli field is $\mathbb{Q}(x, y, z)$, of transcendence degree three.

We show that every four-point cross ratio is transcendental. A cross ratio containing z is a nonconstant fractional linear function of z over $\mathbb{Q}(x, y)$, so is transcendental over that field. For a quadruple not containing z, there are three cases. Three rational anchors and one of x, y, w give a nonconstant fractional linear function of that variable. Two rational anchors and two of x, y, w give a nonconstant function of two independent variables: each pair among

x, y, w is algebraically independent, since its affine change of variables from (x, y) is invertible. Finally, one rational anchor $a \in \{-2, 0, 2\}$ and all of x, y, w have, in one ordering, cross ratio

$$\rho = \frac{(a-y)(x-w)}{(a-w)(x-y)} = \frac{2(a-y)(x+y+2)}{(a+4+x+2y)(x-y)}. \tag{6.1}$$

It is nonconstant: its denominator has x^2 coefficient one, whereas its numerator has x^2 coefficient zero and y^2 coefficient -2. Thus the numerator cannot be a constant multiple of the denominator. Other orderings give one of the six invertible fractional linear transforms of ρ, and remain nonconstant.

The only elements of a pure rational function field over $\mathbb{Q}$ algebraic over $\mathbb{Q}$ are its rational constants. To check this directly, a separable polynomial relation would make all partial derivatives zero. For a reduced fraction P/Q, the equality $\partial_j(P/Q) = 0$ gives $P \mid \partial_j P$ when $P \neq 0$; degree in the jth variable forces $\partial_j P = 0$, and then $\partial_j Q = 0$. Characteristic zero and repetition in all variables make both constant. None of the preceding cross ratios is therefore algebraic. Corollary B, necessary direction, proves non-Ramsey-ness. That direction was established without either direction of OpenAI's Theorem 1.1. □

6.1 A direct finite obstruction

There is also a short exact certificate that uses the second-order node equations directly, independently of the common-subspace classification. For this family the parameter Jacobian with respect to (x, y, z) is

$$J = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & -2 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Writing $a_j(t) = \sum_{r=0}^{2} c_{jr} t^r$ in (3.7), the nine coefficients must satisfy the following twelve linear equations:

$$\sum_i \gamma_i t_i^l \big(J_{ij} a_k(t_i) + J_{ik} a_j(t_i)\big) = 4 \sum_i \gamma_i t_i^l J_{ij} J_{ik}, \quad l = 0, 1, \quad 1 \le j \le k \le 3. \tag{6.2}$$

For $j = k$, this is twice the coefficient equation for δ_j^2; for $j < k$ it is the coefficient equation for $\delta_j \delta_k$. Thus the factor four is uniform.

Order the coefficient columns as $(c_{10}, c_{11}, c_{12}, c_{20}, c_{21}, c_{22}, c_{30}, c_{31}, c_{32})$, and order rows first by l, then by $(j, k) = (1,1), (1,2), (1,3), (2,2), (2,3), (3,3)$. Let M be the first ten rows of the augmented matrix. At the expression test $(x, y, z) = (-3, -1, 3)$, the seven nodes are distinct, so every denominator $H'(t_i)$ is nonzero. The matrix $720M$ is

$$\begin{pmatrix} -28 & -36 & -12 & 0 & 0 & 0 & 0 & 0 & 0 & 64 \\ -15 & -45 & -15 & -14 & -18 & -6 & 0 & 0 & 0 & 120 \\ 1 & 3 & 9 & 0 & 0 & 0 & -14 & -18 & -6 & 0 \\ 0 & 0 & 0 & -30 & -90 & -30 & 0 & 0 & 0 & 300 \\ 0 & 0 & 0 & 1 & 3 & 9 & -15 & -45 & -15 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 6 & 18 & 4 \\ -36 & -12 & -84 & 0 & 0 & 0 & 0 & 0 & 0 & 48 \\ -45 & -15 & -45 & -18 & -6 & -42 & 0 & 0 & 0 & 120 \\ 3 & 9 & 27 & 0 & 0 & 0 & -18 & -6 & -42 & 0 \\ 0 & 0 & 0 & -90 & -30 & -90 & 0 & 0 & 0 & 180 \end{pmatrix}.$$

Fraction elimination and integer Bareiss elimination give

$$\det(720M) = 1223059046400000, \qquad \det M = \frac{1}{30611001600000}. \tag{6.3}$$

Both calculations and a left-null-vector certificate are in `second_jet_certificate.py`. The nonzero test value proves that this augmented minor is a nonzero rational function of x, y, z. At every algebraically independent triple it stays nonzero, since its nonzero numerator cannot vanish. Nine coefficient columns cannot then span the right-hand side of the ten-row system. This rules out a tensor certificate; Proposition 2.2 gives an avoiding colouring with 15^7 colours in every dimension, without using either direction of OpenAI's classification.

The rational test is a test of finitely many rational expressions, not a specialization homomorphism from $\mathbb{Q}(x, y, z)$ to $\mathbb{Q}$. In particular it makes no non-Ramsey assertion about the rational seven-point configuration used in the determinant computation.

6.2 Why the derivation-certificate question is different

The discussion in [4, Appendix E, after Theorem A.8] concerns an identity

$$\sum_i \lambda_i \|D(a_i')\|^2 = C \neq 0$$

for one derivation D, fixed real scalar weights λ_i, and every congruent copy in $\mathbb{R}^3$. The absence of such an identity is a restriction on this particular non-Ramsey certificate method. It does not imply tensor-certificate existence. The joint second-order conditions (6.2) involve mixed coefficients from three independent parameter directions and are inconsistent. Thus the determinant obstruction and the reported failure of the single-derivation method have compatible conclusions.

7 An elliptic boundary example

We prove Theorem D by rational identities. No elliptic-curve group law is needed. Work first in the function field

$$F = \mathbb{Q}(x, y), \qquad y^2 = x^3 - 2,$$

where x is transcendental. Define the rational map

$$\begin{aligned} x' &= \frac{\mathcal{A}(x,y)}{(x-3)^2}, \quad \mathcal{A}(x,y) = 3x^2 + 9x - 4 - 10y, \\ y' &= \frac{\mathcal{B}(x,y)}{(x-3)^3}, \quad \mathcal{B}(x,y) = 5x^3 + 45x^2 - 40 - (27x + 19)y. \end{aligned} \tag{7.1}$$

Write (x_i, y_i) for its iterates, starting with $(x_0, y_0) = (x, y)$. The circle parameters will be $x_0, \ldots, x_6$.

7.1 A finite rational test and real embeddings

The polynomial identity

$$\mathcal{B}(x,y)^2 - \mathcal{A}(x,y)^3 + 2(x-3)^6 = (y^2 - x^3 + 2)(-171x^2 - 1674x + 1000y + 1561) \tag{7.2}$$

shows that the map preserves the curve wherever defined. To check that the required denominators are nonzero as functions, and the iterates distinct and nonconstant, use the curve point $P_0 = (129/100, -383/1000)$. Eight successive first coordinates are the following fractions, denoted r_i:

i	$r_i = x_i(P_0)$
0	$\frac{129}{100}$
1	$\frac{164323}{29241}$
2	$\frac{2340922881}{58675600}$
3	$\frac{307326105747363}{160280942564521}$
4	$\frac{794845361623184880769}{5131273100073606144900}$
5	$\frac{4968031750452922778611893 7923}{345851910470261667904471 9441}$
6	$\frac{3003708872463045080338203553 8503505921}{3010683982898763071786842993779918400}$
7	$\frac{1823868975684837630896890992509406508726006192 03}{1275723963353050497405470386463457417988593644 01}$

Exact cross multiplication gives

$$r_0 < r_7 < r_4 < r_3 < r_1 < r_6 < r_5 < r_2, \qquad r_i \neq 3 \quad (0 \le i \le 7). \tag{7.3}$$

The fractions and comparisons are checked by `elliptic.py`. Each iterate is therefore defined as a rational function, and the first seven x_i are distinct. If P_1 is the image of P_0, then $x_i(P_1) = r_{i+1} \neq r_i$ for $0 \le i \le 6$, so each x_i is nonconstant.

The rational function $x^3 - 2$ is not a square in $\mathbb{Q}(x)$, by its odd order at infinity. For every transcendental real $X > \sqrt[3]{2}$ and either sign $Y = \pm\sqrt{X^3 - 2}$, substitution defines an embedding of F into $\mathbb{R}$. Indeed $\mathbb{Q}(x) \simeq \mathbb{Q}(X)$, and a relation $Y \in \mathbb{Q}(X)$ would make $x^3 - 2$ a square after clearing denominators. The quadratic field extension therefore embeds injectively. Every nonzero denominator and difference of functions just checked remains nonzero at (X, Y). The actual circle coordinate field is the embedded copy of F: the coordinates recover $X_0 = X$ and X_1, and

$$Y = \frac{3X^2 + 9X - 4 - (X - 3)^2 X_1}{10}.$$

7.2 The tensor ring in this example is a domain

This property is needed only for the following construction. Put $K = \mathbb{Q}(x, u)$. The element $x^3 - 2$ is not a square in K. In $K(y)$, a proposed square root of $u^3 - 2$ has the form $a + by$, $a, b \in K$. Comparing the y coefficient in its square gives $ab = 0$. If $b = 0$, it would make $u^3 - 2$ a square in K; if $a = 0$, it would make $(u^3 - 2)/(x^3 - 2)$ a square in K. Both have odd order at $u = \infty$, so neither is possible. Consequently $K(y, v)$ with $y^2 = x^3 - 2$, $v^2 = u^3 - 2$ has degree four over K.

The ring

$$A_0 = \mathbb{Q}[x, u, y, v]/(y^2 - x^3 + 2, v^2 - u^3 + 2)$$

is free over $\mathbb{Q}[x, u]$ with basis $1, y, v, yv$. These elements are independent in the degree-four field just constructed, so A_0 injects into it and is a domain. The tensor ring $\mathcal{B}_F = F \otimes_{\mathbb{Q}} F$ is the localization of A_0 at the nonzero pure left and pure right elements. Thus it too is a domain. This argument is specific to this function field; the preceding classification and two-group certificate assumed no such property.

7.3 A simultaneous invariant

For the two copies of the function field write

$$Z = (x - u)^2, \qquad S = xu(x + u) - 4 + 2yv, \qquad d = S/Z.$$

The last quotient is taken only in the fraction field of $\mathcal{B}_F$. The same map (7.1) on both pairs (x, y) and (u, v) preserves d. Here is a polynomial verification. Write

$$D = x - 3, \quad E = u - 3, \quad a = \mathcal{A}(x, y), \quad a' = \mathcal{A}(u, v), \quad b = \mathcal{B}(x, y), \quad b' = \mathcal{B}(u, v),$$

and put

$$\begin{aligned} M &= aa'(aE^2 + a'D^2) - 4D^4E^4 + 2bb'DE, \\ \mathcal{C} &= M(x - u)^2 - S(aE^2 - a'D^2)^2. \end{aligned} \tag{7.4}$$

Division by the two monic curve equations gives

$$\mathcal{C} = Q_1(y^2 - x^3 + 2) + Q_2(v^2 - u^3 + 2). \tag{7.5}$$

The quotients are displayed in Section 8, and `elliptic.py` performs the divisions with exact coefficients. The transformed S and Z have common denominator D^4E^4, with numerators M and $(aE^2 - a'D^2)^2$, respectively. Thus (7.5) proves invariance of d.

All the needed differences $L(x_i) - R(x_i)$ are nonzero. In fact x_i is nonconstant and hence not rational, so there is a $\mathbb{Q}$-linear functional $\lambda : F \to \mathbb{Q}$ with $\lambda(1) = 0$ and $\lambda(x_i) = 1$. Applying $\lambda \otimes \mathrm{id}$ to a proposed equality $x_i \otimes 1 = 1 \otimes x_i$ gives $1 = 0$. The denominators in every d_i are consequently nonzero. The invariant gives $d_i = d$ along all seven simultaneous iterates.

Set

$$K(t, w) = tw(t + w) - 4, \qquad C(t, w) = t^2w^2 + 8(t + w).$$

The further polynomial identity

$$K(t, w)^2 - (t - w)^2C(t, w) = 4(t^3 - 2)(w^3 - 2) \tag{7.6}$$

implies, for $t_i = L(x_i)$ and $u_i = R(x_i)$,

$$(t_i - u_i)^2d^2 - 2K(t_i, u_i)d + C(t_i, u_i) = 0. \tag{7.7}$$

Indeed $S_i = K(t_i, u_i) + 2L(y_i)R(y_i)$ and $d = S_i/(t_i - u_i)^2$; expanding (7.6) proves (7.7).

7.4 A certificate with only pure-factor denominators

In $\mathcal{B}_F[t, w]$, define

$$G(t, w) = -2L((4y^2)^{-2})\big[(t - w)^2S^2 - 2K(t, w)ZS + C(t, w)Z^2\big]. \tag{7.8}$$

This has degree at most two in each variable. Its only inverse is a pure left factor, since $4y^2 \neq 0$ in F. Multiplying (7.7) by Z^2 proves that (7.8) vanishes at each tensor node, first in the fraction field and therefore in the domain $\mathcal{B}_F$. On multiplying coefficients, $m(Z) = 0$ and $m(S) = 4y^2$, so

$$m(G(t, w)) = -2(t - w)^2.$$

Equation (2.3) gives a tensor matrix with multiplied value H_0 and zero node evaluations. The seven points are distinct, hence span the plane. OpenAI's Theorem 1.1, sufficient direction, proves the positive Ramsey assertion of Theorem D.

7.5 All thirty-five cross ratios are transcendental

For $0 \le a < b < c < d \le 6$, put

$$\rho_{abcd} = \frac{(x_a - x_c)(x_b - x_d)}{(x_a - x_d)(x_b - x_c)} \in F.$$

At P_0 and P_1 its values are respectively $\mathrm{cr}(r_a, r_b; r_c, r_d)$ and $\mathrm{cr}(r_{a+1}, r_{b+1}; r_{c+1}, r_{d+1})$, where here the notation denotes the corresponding difference formula. All differences in the denominators are nonzero by (7.3). Computing first value minus second value, the exact signs in lexicographic order of (a, b, c, d) are

$$\texttt{-+-++-++---+-++++--++++--+++}. \tag{7.9}$$

These are 35 nonzero signs. The rational table, the difference formula, and (7.9) form a finite certificate; `elliptic.py` enumerates it with `fractions.Fraction` and saves every exact difference.

If ρ_{abcd} were algebraic over $\mathbb{Q}$, its irreducible polynomial would have a rational root upon evaluation at P_0. This evaluation is legitimate for that finite expression: every denominator used is nonzero at P_0. The irreducible polynomial would therefore be linear and ρ_{abcd} a rational constant, contradicting its different value at P_1. Thus it is transcendental in F, and injectivity of the real embedding preserves this fact. All other orderings are invertible fractional linear transforms, so all four-point cross ratios are transcendental.

Finally $M(A) \subset F$ and $\mathrm{trdeg}_{\mathbb{Q}} F = 1$, while $M(A)$ contains a transcendental cross ratio. Hence $\mathrm{trdeg}_{\mathbb{Q}} M(A) = 1$. This completes the proof of Theorem D, with the positive Ramsey conclusion depending exactly on OpenAI's sufficient direction as specified.

8 Exact identity certificates

The ancillary directory contains the five scripts cited in this note, together with a validator that compares the certificates printed here with their output. They use Python 3, its standard library, and SymPy; all numerical tests are exact rational or integer calculations. The README gives serial commands with the resource bounds. In particular:

- `identities.py` checks the homogeneous determinant and circle-kernel identities and the barycentric moment formulas.
- `structure.py` checks the 64 seven-node sign spaces, their projection dimensions, and an algebraic-diagonal certificate.
- `eight_factory.py` checks the 128 eight-node sign spaces and the universal two-group graph identity.
- `second_jet_certificate.py` checks (6.3) by two determinant algorithms and saves an exact left-null-vector certificate.
- `elliptic.py` checks the polynomial identities (7.2), (7.5), and (7.6); the rational orbit; and every cross-ratio difference in (7.9). It also performs 21 rational tests of the simultaneous invariant.

Rational expression tests prove the specified nonvanishing and nonconstancy claims. The general statements about all isotropic spaces are proved in Section 4, and the invariant is proved by its polynomial identity, rather than by the 21 sample tests.

For an explicit certificate of (7.5), the two polynomial quotients are

$$\begin{aligned} Q_1 = -20(u-3)^2\big(&5u^4x - 15u^4 + 5u^3x^2 - 45u^3 - 54u^2vx + 10u^2vy + 112u^2v \\ &- 45u^2x^2 + 135u^2x + 54uvx^2 - 100uvx - 60uvy + 114uv \\ &- 40ux + 120u - 20v^2x^2 + 120v^2x - 180v^2 - 12vx^2 \\ &- 114vx + 90vy + 20x^2 - 180\big), \end{aligned}$$

and

$$\begin{aligned} Q_2 = -20(x-3)^2\big(&-15u^2x^3 - 45u^2x^2 + 54u^2xy - 12u^2y + 60u^2 \\ &+ 5ux^4 + 120ux^3 - 54ux^2y + 135ux^2 - 100uxy - 40ux \\ &- 114uy - 240u + 10vx^2y - 60vxy + 90vy - 15x^4 \\ &- 225x^3 + 112x^2y + 114xy + 120x + 180\big). \end{aligned}$$

Substitution into (7.5) and expansion gives an identity in $\mathbb{Q}[x, u, y, v]$. The script obtains the same quotients by successive polynomial division in y and v and verifies the zero remainder.

References

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[2] OpenAI. A classification of finite Euclidean Ramsey configurations. OpenAI Math Release, result 172, September 23, 2026. Theorem 1.1. Pinned repository commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. `https://github.com/openai/math/blob/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/A-classification-of-finite-Euclidean-Ramsey-configurations-September-23-2026/paper.pdf`.

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https://github.com/wstrinz/spherical-ramsey/blob/09f88943111035c1764169
cfd96bde7340aff687/note/spherical-ramsey.tex.

Use of AI

The proofs and the text of this note were produced with AI systems (GPT-6.1 Sol, with editing by Claude Opus 5.5) under the author's direction. They were checked by independent reviews with AI models (GPT-6.1 Sol and Gemini 4 Argon) and, for the explicit identities and finite arithmetic certificates, by the exact computations in the ancillary files. No human expert has reviewed this note. The author takes responsibility for the content. Every positive Ramsey conclusion depends on the sufficient direction of OpenAI's classification, Theorem 1.1 of result 172, at commit adc7f1241b42e322a6451854ab7e4b4c146bf78a [2]. The negative conclusions are justified by the independent tensor-necessity and colouring argument in Section 2, so do not require either direction of that theorem.